# SIM-01 Closure and SIM-02A Review Transmittal

**Ulmo / Picket, PCK-SIM-001: SIM-01 documentation corrections and SIM-02A sensor, noise and error modeling**

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| To | Author of the SIM-01 technical review and SIM-02 authorization |
| Subject | SIM-01 closure (report Revision A) and SIM-02A numerical results for review |
| Date | 2026-11-07 |
| Repository | `keithatulmo/picket-sim`, branch `claude/beautiful-darwin-vbeoue` |
| SIM-02A run of record | `20261107T214846Z_c56c0048_31da01b` (code commit `31da01b`, clean tree) |
| Prepared by | Claude Code (AI-generated engineering work product). Requires review by a qualified acoustic engineer before any of it informs requirements. |
| Requested action | Confirm SIM-01 closure; review the SIM-02A numerical results; decide the items in Section 6 before SIM-02B is authorized |

## 1. SIM-01 documentation corrections (review Section 1)

All five corrections are made without altering any experimental result, and recorded in ADR-0008, `results/sim01/ERRATA.md` and the report's revision history. The Rev 0 review package is retained unchanged.

| # | Correction | Where |
|---|---|---|
| 1 | 6.02 dB qualified as the white-noise gain of four equally weighted elements, not an absolute bound | SIM-01 report Rev A Sections 0, 4.2, 5.1; SIM-01 transmittal Rev A |
| 2 | f_HP described as a -3 dB discrimination criterion, not a limit on direction information | Report Rev A Sections 0, 4.1, 5.1 to 5.3, 8; `docs/metrics.md` |
| 3 | Median-steering bands separated from bands guaranteed across the steering set | `reports/SIM-01_band_definitions.md` (derived from the existing tables by `tools/sim01_band_definitions.py`); report Rev A Sections 0, 4.1, 5.2 |
| 4 | Exact narrowband grating lobes separated from broadband direction finding | Report Rev A Sections 0, 4.1, 4.4, 5.4, 7, 8 |
| 5 | Pareto conclusions conditional; no elimination of hardware candidates | Report Rev A Sections 0, 6, 7, 8 |

Correction 3 changes the headline band ratios. Over the horizon steering set the bands guaranteed for every direction are 1.77 (lines), 1.28 (A3) and 1.06 (A4). The median-direction bands are 3.54, 1.54 and 1.36. The Rev 0 values (3.03, 1.44, 1.18) used a mixed convention.

## 2. SIM-02A deliverables (review Section 5)

| # | Deliverable | Status | Evidence |
|---|---|---|---|
| 1 | Updated technical implementation plan | Delivered | `docs/SIM-02A_plan.md`; ADR-0009 to ADR-0013 (written before implementation) |
| 2 | SIM-02A source, configuration, automated tests | Delivered | `src/picket_sim/{params,sensor,noisefields,errors,stats,rag,avs}.py`, `experiments/sim02a.py`; `configs/sim02a/`; 166 tests passing |
| 3 | Sensor, noise and error model documentation | Delivered | `docs/sensor_noise_error_models.md` |

| # | Deliverable | Status | Evidence |
|---|---|---|---|
| 4 | Comparative results, retained and new candidates | Delivered | SIM-02A report Sections 2 and 4; run tables and figures |
| 5 | Sensitivity and uncertainty analysis | Delivered | Report Sections 4.2 (eta sweep), 4.4 (Monte Carlo with 95 % intervals, one-at-a-time sweeps), 4.5 (-3/-6/-10 dB), 4.6 (tolerance scales) |
| 6 | Updated limitations and assumption register | Delivered | `docs/assumptions.md` (AS-11 to AS-19); report Section 6 |
| 7 | Reproducible engineering report | Delivered | `reports/SIM-02A_engineering_report.md` / `.pdf`; `make sim02a` |
| 8 | Recommendations for SIM-02B | Delivered | Report Section 7 |

Additional item: acoustic vector sensor feasibility assessment (D05), `docs/avs_feasibility.md`. SIM-02B work has not started.

## 3. Verification requirements (review Section 4)

| Requirement | Evidence |
|---|---|
| Positive-semidefinite noise covariance | `check_covariance` on every coherence and total covariance (raises); `test_coherence_models_are_psd_with_unit_diagonal`, `test_invalid_covariances_and_fields_fail_visibly` |
| Correct noise normalization | Unit-diagonal check on every field; gain limits self-checked each run (white field 10 log10 N; isotropic field equals SIM-01 DI, agreement 5e-15 dB) |
| Array gain versus detection performance | Gain reported only as narrowband SNR ratios relative to A0; detection deferred to SIM-02B (ADR-0011) |
| Calibration and timing-error propagation | Closed forms for signal loss and null filling verified by Monte Carlo (5 error types in tests; 1,664 cases in the run, largest deviation 3.33 standard errors) |
| Broadband processing assumptions | None made in SIM-02A; narrowband scope stated in every result |
| Statistical uncertainty in Monte Carlo results | 95 % intervals on means, quantiles and proportions; coverage tested; order-independent seeded streams |
| Consistent comparison of geometries | Footprint classes C (0.45 m) and E (0.90 m); identical steering, noise and error models for every candidate |
| Fail visibly | Provenance, unit, Nyquist, bandwidth, PSD, convergence and normalization violations raise; undefined statistics reported as n/a with reasons; no clipping in error paths |

## 4. Errors found and corrected during SIM-02A

Recorded in `results/sim02a/ERRATA.md`; both earlier runs are retained unchanged.

- Run 1: line-array bearing statistics were a tie-breaking artifact. The scanned response of a line is exactly symmetric in azimuth, and `argmax` picked the mirror. Cut statistics were also reported where the error-free pattern has no half-power discrimination.
- Run 2: the first fix classified outright gross errors as ties.
- Run of record: structural ambiguity is decided on the error-free cut (outside lobe within 1 dB of the mainlobe). Gross errors are counted only where no structural ambiguity exists.

- Before any run, the assumed sensor scenario S1 was corrected to be physically consistent. Its element sensitivity had been set as if it included preamplifier gain, while the same value was used to refer the element's thermal noise.

Every table other than the Monte Carlo bearing columns is byte-identical across the three runs.

## 5. Principal SIM-02A findings

1. Realized gain is governed by the spatial coherence of the dominant noise. In diffuse, ambient-limited noise, arrays up to 1.07 m gain at most 0.07 dB over A0 from 20 to 200 Hz. In sensor-limited or short-correlation noise the same arrays approach 10 log10 N.
2. Under the assumed sensors, ambient noise dominates almost everywhere. B4 is not acquirable by the low-cost scenario, and 16-bit quantization limits the mid scenario in B3 and B4.
3. At equal footprint, 6- and 8-element circles widen the -3 dB guaranteed band from 1.28 (square) to 2.53 and 3.78, but only to 1.28 and 1.32 under -10 dB.
4. Four-element volumetric envelopes have no guaranteed band at -6 or -10 dB (narrowband).
5. Bearing statistics are stricter than signal loss. For example, 10 us timing error at 15 kHz causes gross azimuth errors in 65 % of trials for the 8-element compact circle, at a 4.4 dB signal loss.
6. Line arrays stay structurally ambiguous left/right under any channel errors.
7. Pareto dominance changes with the provisional tolerances; no candidate is eliminated.

All sensor, ambient, host and error levels are assumed. Ratio-based results transfer; scenario-specific numbers do not.

## 6. Decisions requested before SIM-02B

1. **Inputs with provenance.** Which published or measured sources to adopt for hydrophone and preamplifier self-noise, ambient noise, host noise and vibration, and flow-noise correlation (report Section 7, item 4). SIM-02B results cannot inform requirements without them.
2. **SIM-02B candidate subset and priorities** (report Section 7, item 5).
3. **Detection experiment design:** false-alarm probabilities, integration times and the signal classes to carry forward from D01.
4. **Synchronization model** for distributed modules (timing dominates at high frequency).
5. **Whether to fund the acoustic vector sensor data collection** in `docs/avs_feasibility.md`.
6. **The structural-ambiguity level** (-1 dB) used for bearing statistics, and the Monte Carlo steering set (4 azimuths), if a different convention is preferred.

## 7. Suggested review path (about 45 minutes)

1. This transmittal, Sections 5 and 6.
2. SIM-02A report Section 0 (findings), Figures 2, 5 and 6, and Section 7 (SIM-02B recommendations).
3. `tables/parameters_by_provenance.md`, to confirm that nothing is presented as measured.
4. SIM-01 report Rev A Section 0 and `reports/SIM-01_band_definitions.md`, for the closure corrections.
5. Optional: reproduce with `make sim02a` (about 6 minutes including the test suite) and compare against `SHA256SUMS.txt`.